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ABOUT

Computer Engineer with experience in hardware design and embedded systems development.

EXPERIENCE

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| Millennium Space Systems | El Segundo, CA | August 2024 – Present |
| Designed and implemented a Verilog module supporting bi-directional translation between Ethernet and a proprietary network protocol. Included calculating and verifying IP and UDP checksums. Worked on integrating custom IP into Xilinx Block Diagrams | | |
| Pacific Northwest National Laboratory | Remote – Buffalo, NY | May – August 2022, 2023 |
| Extended the COMET compiler, focusing on end-to-end compilation from a DSL through execution on heterogeneous compute including FPGA, GPU, and CPU. | | |
| Annapolis Microsystems | Annapolis, MD | January – May 2022 |
| Expanded coverage of digital logic synthesis and simulation regression testing system. Triaged timing failures in large HDL projects. | | |
| Pacific Northwest National Laboratory | Remote – Buffalo, NY | July – September 2021 |
| Worked on establishing a pipeline for compiling Python code into Verilog using the MLIR project, IREE, and Tensorflow. | | |

SKILLS

Programming Languages: Python, VHDL, Verilog, C, C++, TCL
Tools: Linux, Quartus, ModelSim, Microchip Libero, Xilinx Vivado, Git
Hardware: Oscilloscope, Digital Multimeter, Waveform Generator, ILA Debugging

PROJECTS

- **Digital System Design II Lab:** Designed and verified a MIPS processor in VHDL using Xilinx Vivado. Independently implemented jump/branch instructions and some exception handling.
- **Open-Source Contributions:** Contributed Java code and Android UI XML to the open-source MTG Familiar Android app. Functionality was added to allow parsing Magic: The Gathering cards using regex. Full properties for the cards were retrieved from a SQL database and used to display metrics.
- **Senior Design:** Working on an autonomous chess board that can move pieces without user interaction. Uses an electromagnet on a gantry for movement and RFID for piece identification. Controlled by a Raspberry Pi running Stockfish.

EDUCATION

ROCHESTER INSTITUTE OF TECHNOLOGY, Rochester, NY
Bachelor/Master of Science in Computer Engineering, May 2024
GPA: 3.43/4.00
Awards: Dean's List Fall 2019 and 2020; RIT Founder's Merit Scholarship; 2019 National Merit Scholarship semi-finalist; AP + PLTW Student Achievement in Engineering

INTERESTS

- Magic: The Gathering
- 3D Printing